United States Patent Application Publication 20250287140
Kind Code A1
Publication Date September 11, 2025
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WOOFER SYSTEM, ACOUSTIC SYSTEM, SEAT, AND MOBILE OBJECT

Abstract

A woofer system that is to be attached to a seat includes a woofer and a sound path element that forms a sound path that leads a sound output by the woofer. The sound path element includes an opening arranged close to an ear of a listener who sits on the seat.

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Appl. No.: **19/062955**

Filed: **February 25, 2025**

Foreign Application Priority Data

JP	2024-032683	Mar. 05, 2024

Publication Classification

Int. Cl.: **H04R1/28** (20060101); **H04R1/02** (20060101); **H04R1/26** (20060101); **H04R1/34** (20060101)

U.S. Cl.:

CPC **H04R1/2803** (20130101); **H04R1/025** (20130101); **H04R1/26** (20130101); **H04R1/345** (20130101); H04R2499/13 (20130101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application is based on and claims priority of Japanese Patent Application No. 2024-032683 filed on Mar. 5, 2024.

FIELD

[0002] The present disclosure relates to a woofer system attached to a seat, an acoustic system including the woofer system, a seat into which the woofer system is incorporated, and a mobile object including the woofer system.

BACKGROUND

[0003] Conventionally, a woofer that produces sounds in a relatively low frequency sound range tends to be arranged at a position apart from listeners. In this case, it is necessary to increase the sound level of sounds produced from the woofer to allow the listeners to listen to the sounds at an adequate sound level. However, the sounds increased in the sound level may be noise to non-listeners. Moreover, when the sounds are produced in, for example, a listening room, sounds leaked outside may be considered to be noise.

[0004] For example, Patent Literature (PTL) 1 discloses a technique for arranging a woofer near a listener by attaching the woofer to a seat that the listener sits on.

CITATION LIST

Patent Literature

[0005] PTL 1: Japanese Unexamined Patent Application Publication No. 2006-020163

SUMMARY

[0006] However, the woofer according to PTL 1 can be improved upon.

[0007] In view of this, the present disclosure provides a woofer system, an acoustic system, a seat, and a mobile object which are capable of improving upon the above related art.

[0008] A woofer system according to one aspect of the present disclosure is a woofer system to be attached to a seat. The woofer system includes a woofer, and a sound path element that forms a sound path that leads a sound output by the woofer. The sound path element includes an opening arranged close to an ear of a listener who sits on the seat.

[0009] An acoustic system according to one aspect of the present disclosure includes: a woofer system that is to be attached to a seat and includes a woofer and a sound path element that forms a sound path that leads a sound output by the woofer, where the sound path element includes an opening arranged close to an ear of a listener who sits on the seat; and a loudspeaker that is attached to the seat in a vicinity of the ear of the listener who sits on the seat, and reproduces an acoustic signal in a mid-to-high frequency sound range that is a range including frequencies higher than frequencies at which the woofer reproduces an acoustic signal.

[0010] A seat according to one aspect of the present disclosure includes a woofer, and a sound path element that forms a sound path that leads a sound output by the woofer. The sound path element includes an opening arranged close to an ear of a listener who sits on the seat.

[0011] A mobile object according to one aspect of the present disclosure includes at least one woofer system that is to be attached to a seat and includes a woofer and a sound path element that forms a sound path that leads a sound output by the woofer, where the sound path element includes an opening arranged close to an ear of a listener who sits on the seat or a seat that includes a woofer and a sound path element that forms a sound path that leads a sound output by the woofer, where the sound path element includes an opening arranged close to an ear of a listener who sits on the seat.

[0012] A woofer system, an acoustic system, a seat, and a mobile object according to one aspect of the present disclosure are capable of improving upon the above related art.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] These and other advantages and features of the present disclosure will become apparent from the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present disclosure.

[0014] FIG. **1** is a perspective front view of an acoustic system attached to a seat.

[0015] FIG. **2** is a perspective back view of the acoustic system attached to the seat.

[0016] FIG. **3** is a perspective front view of a woofer system.

[0017] FIG. **4** is a perspective front view of the inside of the woofer system.

[0018] FIG. **5** is a perspective back view of the woofer system.

[0019] FIG. **6** is a perspective back view of the inside of the woofer system.

[0020] FIG. **7** is a perspective front view of the inside of a woofer system according to Example 1.

[0021] FIG. **8** is a perspective back view of a woofer system according to Example 2.

[0022] FIG. **9** is a perspective front view of the inside of the woofer system according to Example 2.

[0023] FIG. **10** is a diagram illustrating a mobile object that includes the acoustic system.

[0024] FIG. **11** is a perspective back view of a seat into which the woofer system is incorporated.

DESCRIPTION OF EMBODIMENT

[0025] Hereinafter, an embodiment of a woofer system, an acoustic system, and a mobile object according to the present disclosure will be described with reference to the drawings. Note that the following embodiment is a mere example presented to describe the present disclosure, and thus is not intended to limit the present disclosure. For example, the shapes, structures, materials, elements, relative positional relationships, connection states, numerical values, mathematical expressions, details of steps in methods, and the orders of the steps, etc. presented in the embodiment below are mere examples, therefore may include contents not described below. Moreover, geometrical expressions, such as parallel and orthogonal, may be used, but these expressions do not denote mathematical strictness. These expressions include substantially allowable errors, deviations, etc. Expressions, such as simultaneous and identical, also include a substantially allowable range of errors, etc.

[0026] The drawings are schematic diagrams on which emphases, omissions, and/or proportional adjustments are performed as appropriate to describe the present disclosure. Accordingly, the drawings are different from the actual shapes, actual positional relationships, and/or actual proportions. In addition, the X axis, Y axis, and Z axis which may be shown in the drawings denote an orthogonal coordinate system optionally set for describing the drawings. In other words, the Z axis is not limited to an axis extended parallel to a vertical direction, and the X axis and the Y axis are not limited to axes present within a horizontal plane.

[0027] The following may comprehensively describe a plurality of inventions as one embodiment. Moreover, some contents stated below are described as optional elements relating to the present disclosure.

[0028] FIG. **1** is a perspective front view of acoustic system **400** attached to seat **200**. FIG. **2** is a perspective back view of acoustic system **400** attached to seat **200**. It should be noted that although seat **200** and woofer system **100** are simply illustrated by extensive use of straight lines in the drawings, seat **200** may have complicated curved portions, and woofer system **100** may have curved portions that fit the shape of seat **200**.

[0029] Acoustic system **400** includes woofer system **100** attached to seat **200** and loudspeakers **410** provided in the vicinity of the ears of a lister who sits on seat **200**, and reproduces acoustic signals in a mid-to-high frequency sound range that is a range including frequencies higher than frequencies at which woofer **110** reproduces acoustic signals. Loudspeaker **410** may be at least one of a tweeter or a squawker. An acoustic signal in the mid-to-high frequency sound range is a signal whose maximum peak of sound pressure is present in a frequency range above 300 Hz.

[0030] In the present embodiment, loudspeakers **410** are attached to headrest **230** of seat **200**. Acoustic system **400** includes a plurality of loudspeakers **410**, and two loudspeakers **410** are arranged in the vicinity of respective ears of a listener who sits on seat **200**. Two openings **160** and the two loudspeakers **410** are respectively arranged adjacent to each other.

[0031] As long as seat **200** is a seat that a listener sits on, materials, shapes, etc. of seat **200** are not limited. For example, when seat **200** is provided in an automobile, seat **200** may be, for example, the driver's seat or the passenger's seat next to the driver's seat. Moreover, when seat **200** is arranged in a living room, seat **200** may be a sofa. In the present embodiment, seat **200** includes seat component **210**, backrest component **220**, and headrest **230**. Seat component **210** is one of structural components of seat **200** which supports the buttocks and thighs of a listener who sits on seat **200**. Backrest component **220** is one of the structural components of seat **200** which supports the back of a listener who sits on seat **200**. Headrest **230** is one of the structural components of seat **200** which is arranged at the back of the head of a listener who sits on seat **200**. Seat component **210**, backrest component **220**, and headrest **230** each include, for example, an outer skin component made of leather (including synthetic leather), resin, fabric, etc. and a cushion component covered by the outer skin component.

[0032] FIG. **3** is a perspective front view of woofer system **100**. FIG. **4** is a perspective front view of an inside of woofer system **100**. FIG. **5** is a perspective back view of woofer system **100**. FIG. **6** is a perspective back view of the inside of the woofer system. Woofer system **100** is a device that is attached to seat **200**, and emits sounds close to the ears of a listener who sits on seat **200**. Woofer system **100** may be integrally attached to seat **200** or may be an individual element separable from seat **200**. In the present embodiment, woofer system **100** is a device attachable to and detachable from the back surface of backrest component **220** of seat **200**, and includes woofer **110** and sound path element **150**.

[0033] Woofer **110** is a loudspeaker that reproduces audio signals in a low frequency sound range. An acoustic signal in the low frequency sound range is a signal whose maximum peak of sound pressure is present in a frequency range of from 20 Hz to 300 Hz (particularly, at most 100 Hz). The shape and structure of woofer **110** are not limited. As an example of woofer **110**, a loudspeaker including a diaphragm, a voice coil, a magnetic circuit, and a frame can be presented. In the present embodiment, woofer **110** is a loudspeaker that includes a cone-shaped diaphragm. In the present embodiment, it should be noted that, in a winding axis direction of a voice coil, a side on which the voice coil is arranged relative to the diaphragm of woofer **110** may be stated as back, and the side opposite thereof may be stated as front.

[0034] In the present embodiment, woofer **110** is attached to, as illustrated in FIG. **4** and FIG. **6**, enclosure **111** that separates sounds output from the front of woofer **110** and sounds output from the back of woofer **110**.

[0035] Sound path element **150** is a component that forms a sound path that leads sounds emitted from woofer **110** attached to seat **200** to a desired position. Sound path element **150** includes openings **160** arranged close to the ears of a listener who sits on seat **200**. In the present embodiment, sound path element **150** includes rear sound path element **151** (see FIG. **6**) that leads sounds output from the back of woofer **110** and front sound path element **152** (see FIG. **4**) that leads sounds output from the front of woofer **110**.

[0036] Rear sound path element **151** is a component that forms a sound path that leads rear sound **301** (arrows in FIG. **6** show the route) output from the back of woofer **110** to rear sound openings **161** arranged close to the ears of a listener who sits on seat **200**. The structure of rear sound path element **151** is not limited, but rear sound path element **151** may have a structure in which the phase of rear sound **301** output from rear sound openings **161** agrees with the phase of front sound **302** output from front sound openings **162**. The present embodiment adopts a structure that increases a desired frequency level by adjusting at least one of the volume of enclosure **111** (may be absent) that communicates with rear sound path element **151**, the cross-section area size of rear sound path element **151**, or the length of rear sound path element **151**. The cross-section area size of rear sound path element **151** is an area size of a space of rear sound path element **151** that is taken in a plane orthogonal to the traveling direction of rear sound **301**.

[0037] Rear sound path element **151** may be a tubular component, and, as rear sound openings **161**, openings of the tubular component may be arranged close to the ears of a listener who sits on seat **200**. In the present embodiment, woofer system **100** includes woofer **110** accommodated inside the box-shaped sound path element **150** and separation component **101** (see FIG. **6**) including enclosure **111** that separates rear sound **301** and front sound **302** inside sound path element **150**. Rear sound path element **151** includes separation component **101**, frame component **103** (see FIG. **6**) that is a portion of the outer shell of sound path element **150**, rear component **104** (see FIG. **5**), and guiding component **102** (see FIG. **6**). Rear sound path element **151** includes two rear sound openings **161** arranged in the vicinity of respective ears of a listener who sits on seat **200**.

[0038] Front sound path element **152** is a component that forms a sound path that leads front sound **302** (arrows in FIG. **4** show the route) output from the front of woofer **110** to front sound openings **162** arranged close to the ears of a listener who sits on seat **200**. The structure of front sound path element **152** is not limited, but may be a structure in which the phase of front sound **302** output from front sound openings **162** agrees with the phase of rear sound **301** output from rear sound openings **161**. The present embodiment adopts a structure that increases a desired frequency level by adjusting at least one of the volume of enclosure **111** (may be absent as in the present embodiment) that communicates with front sound path element **152**, the cross-section area size of front sound path element **152**, or the length of front sound path element **152**. Note that the frequency level increased by rear sound path element **151** and the frequency level increased by front sound path element **152** may be different. The cross-section area size of front sound path element **152** is an area size of a space of front sound path element **152** that is taken in a plane orthogonal to the traveling direction of front sound **302**.

[0039] Front sound path element **152** may be a tubular component, and, as front sound openings **162**, openings of the tubular component may be arranged close to the ears of a listener who sits on seat **200**. In the present embodiment, front sound path element **152** includes separation component **101**, frame component **103** (see FIG. **4**) that is a portion of the outer shell of sound path element **150**, and front component **105** (see FIG. **3**). Front sound path element **152** includes two front sound openings **162** arranged in the vicinity of respective ears of a listener who sits on seat **200**. The two rear sound openings **161** and the two front sound openings **162** are respectively arranged adjacent to each other.

[0040] Note that the present disclosure is not limited to the above-described embodiment. For example, the present disclosure may include different embodiments achieved by combining optional elements described in the present description or by excluding some of the elements described in the present description. Moreover, the present disclosure also includes variations obtained by applying various modifications conceivable to those skilled in the art to the above-described embodiment within the scope not departing from the spirit of the present disclosure, namely, the meaning of wording recited in the claims.

[0041] For example, as illustrated in FIG. **7**, woofer system **100** may include front sound opening **162** arranged close to one of the ears of a listener who sits on seat **200** and rear sound opening **161** arranged close to the other of the ears of the listener.

[0042] Moreover, as illustrated in FIG. **8** and FIG. **9**, woofer system **100** may include rear sound path element **151**, but need not include front sound path element **152**. Rear sound path element **151**

may include rear sound opening **161** arranged close to the right ear of a listener who sits on seat **200** and rear sound opening **161** arranged close to the left ear of the listener.
[0043] In addition, as illustrated in FIG. **10**, one or more woofer systems **100** may be provided in mobile object **420**, and may build acoustic system **400** inside mobile object **420** together with loudspeakers **410** provided in mobile object **420**.
[0044] Moreover, woofer system **100** may be arranged for a plurality of seats inside mobile object **420**.
[0045] Although woofer **110** has been described as being arranged behind backrest component **220**, woofer **110** may be arranged below the bottom surface of seat component **210** or inside seat component **210**, and rear sound path element **151** and front sound path element **152** may be arranged inside backrest component **220**. This allows adoption of large-sized woofer **110**.
[0046] Moreover, as illustrated in FIG. **11**, seat **200** may be a seat into which woofer **110** and sound path element **150** that forms a sound path that leads sounds output by woofer **110** are incorporated, and sound path element **150** may include openings arranged close to the ears of a listener who sits on seat **200**. Furthermore, seat **200** may include loudspeakers **410** that reproduce audio signals in a mid-to-high frequency sound range that is a range including frequencies higher than frequencies at which woofer **110** reproduces acoustic signals.
[0047] Woofer system **100** according to aspect **1** includes woofer **110** to be attached to seat **200**, and sound path element **150** that forms a sound path that leads a sound output by woofer **110**. Sound path element **150** includes opening **160** arranged close to an ear of a listener who sits on seat **200**.
[0048] According to aspect **1**, low frequency sounds can be made close to the ears of a listener as compared to the case where woofer **110** is arranged at a position apart from the listener. Accordingly, the sound level of sounds output by woofer **110** can be decreased, and the sound level of low frequency sounds that non-listeners unwillingly hear can be suppressed. Furthermore, when the sounds are produced in, for example, a listening room, noise produced as a result of the sounds leaked outside can be suppressed. In addition, a reduction in power requirements can be facilitated.
[0049] Woofer system **100** according to aspect **2** includes aspect **1** and enclosure **111** that separates a sound output from the front of woofer **110** and a sound output from the back of woofer **110**. In woofer system **100**, sound path element **150** is rear sound path element **151** including rear sound opening (or rear opening) **161** as opening **160**. Rear sound path element **151** leads rear sound **301** output from the back of woofer **110** to rear sound opening **161**.
[0050] According to aspect **2**, sounds output from the back of woofer **110** can be effectively delivered to a listener. This allows the listener to further experience the sensation of low frequency sounds.
[0051] Woofer system **100** according to aspect **3** includes aspect **2**. In woofer system **100**, a desired frequency level is increased by adjusting at least one of a volume of enclosure **111**, a cross-section area size of rear sound path element **151**, or a length of rear sound path element **151**.
[0052] Woofer system **100** according to aspect **4** includes aspect **1** or aspect **2** and enclosure **111** that separates a sound output from the front of woofer **110** and a sound output from the back of woofer **110**. In woofer system **100**, sound path element **150** is front sound path element **152** including front sound opening (or front opening) **162** as opening **160**. Front sound path element **152** leads a front sound output from the front of woofer **110** to front sound opening **162**.
[0053] According to aspect **4**, front sounds can be delivered close to the ears of a listener regardless of the attachment position of woofer **110**.
[0054] Woofer system **100** according to aspect **5** includes aspect **4**. In woofer system **100**, a desired frequency level is increased by adjusting at least one of a volume of enclosure **111**, a cross-section area size of front sound path element **152**, or a length of front sound path element **152**.
[0055] Woofer system **100** according to aspect **6** includes aspect **4**. In woofer system **100**, a frequency level caused to be increased by a volume of enclosure **111**, a cross-section area size of

rear sound path element **151**, and a length of rear sound path element **151** and a frequency level caused to be increased by a volume of enclosure **111**, a cross-section area size of front sound path element **152**, and a length of front sound path element **152** are different.
[0056] Woofer system **100** according to aspect **7** includes aspect **4**. In woofer system **100**, front sound opening **162** and rear sound opening **161** are arranged adjacent to each other.
[0057] Woofer system **100** according to aspect **8** includes aspect **4**. In woofer system **100**, front sound opening **162** is arranged close to one of ears, and rear sound opening **161** is arranged close to the other of the ears.
[0058] According to aspects **7** and **8**, rear sound **301** and front sound **302** output from woofer **110** can be concurrently delivered close to the ears, and thus allow a listener to effectively hear low frequency sounds.
[0059] Woofer system **100** according to aspect **9** includes aspect **1**. In woofer system **100**, sound path element **150** includes a right-side opening **160** arranged close to the right ear of the listener who sits on seat **200** and a left-side opening **160** arranged close to the left ear of the listener.
[0060] According to claim **9**, low frequency sounds can be delivered close to the ears of a listener while a predetermined space is filled with sounds output from one side of woofer **110**.
[0061] Acoustic system **400** according to aspect **10** includes: woofer system **100** according to any one of aspects **1** to **9**; and a loudspeaker that is attached to the seat in the vicinity of the ear of the listener who sits on the seat, and reproduces an acoustic signal in a mid-to-high frequency sound range that is a range including frequencies higher than frequencies at which the woofer reproduces an acoustic signal.
[0062] According to aspect **10**, as compared to the case where woofer **110** is arranged at a position apart from a listener, low frequency sounds can be made close to the ears of a listener, and sounds in a wide frequency band from a low frequency sound range to a high frequency sound range can be delivered to the listener. Accordingly, the sound level of sounds output by woofer **110** can be decreased, and the sound level of low frequency sounds that non-listeners unwillingly hear can be suppressed. Furthermore, when the sounds are produced in, for example, a listening room, noise produced as a result of sounds leaked outside can be suppressed. In addition, a reduction in power requirements can be facilitated.
[0063] Seat **200** according to aspect **11** includes woofer **110**, and sound path element **150** that forms a sound path that leads a sound output by woofer **110**. In seat **200**, sound path element **150** includes opening **160** arranged close to an ear of a listener who sits on seat **200**.
[0064] According to aspect **11**, low frequency sounds can be made close to the ears of a listener as compared to the case where woofer **110** is arranged at a position apart from the listener. Accordingly, the sound level of sounds output by woofer **110** can be decreased, and the sound level of low frequency sounds that non-listeners unwillingly hear can be suppressed. Furthermore, when the sounds are produced in, for example, a listening room, noise produced as a result of sounds leaked outside can be suppressed. In addition, a reduction in power requirements can be facilitated.
[0065] Mobile object **420** according to aspect **12** includes one or more of woofer systems **100** according to aspects **1** to **9** or one or more of the seat according to aspect **11**.
[0066] According to aspect **12**, low frequency sounds can be made close to the ear of a listener as compared to the case where woofer **110** is arranged at a position apart from the listener. Accordingly, leakage of low frequency sounds outside mobile object **420** can be reduced. Therefore, noise produced as a result of sounds leaked outside can be suppressed. In addition, a reduction in power requirements can be facilitated.
[0067] While the embodiment has been described herein above, it is to be appreciated that various changes in form and detail may be made without departing from the spirit and scope of the present disclosure as presently or hereafter claimed.
Further Information about Technical Background to this Application
[0068] The disclosure of the following patent application including specification, drawings, and

claims are incorporated herein by reference in their entirety: Japanese Patent Application No. 2024-032683 filed on Mar. 5, 2024.

INDUSTRIAL APPLICABILITY

[0069] The present disclosure is applicable to houses and commercial facilities that include seats, and mobile objects, such as automobiles, vessels, aircraft, and construction machinery, which include seats.

Claims

1. A woofer system to be attached to a seat, the woofer system comprising: a woofer; and a sound path element that forms a sound path that leads a sound output by the woofer, wherein the sound path element includes an opening arranged close to an ear of a listener who sits on the seat.
2. The woofer system according to claim 1, further comprising: an enclosure that separates a sound output from front of the woofer and a sound output from back of the woofer, wherein the sound path element is a rear sound path element including a rear opening as the opening, the rear sound path element leading a rear sound output from the back of the woofer to the rear opening.
3. The woofer system according to claim 2, wherein a desired frequency level is increased by adjusting at least one of a volume of the enclosure, a cross-section area size of the rear sound path element, or a length of the rear sound path element.
4. The woofer system according to claim 2, further comprising: an enclosure that separates a sound output from the front of the woofer and a sound output from the back of the woofer, wherein the sound path element is a front sound path element including a front opening as the opening, the front sound path element leading a front sound output from the front of the woofer to the front opening.
5. The woofer system according to claim 4, wherein a desired frequency level is increased by adjusting at least one of a volume of the enclosure, a cross-section area size of the front sound path element, or a length of the front sound path element.
6. The woofer system according to claim 4, wherein a frequency level caused to be increased by a volume of the enclosure, a cross-section area size of the rear sound path element, and a length of the rear sound path element and a frequency level caused to be increased by a volume of the enclosure, a cross-section area size of the front sound path element, and a length of the front sound path element are different.
7. The woofer system according to claim 4, wherein the front opening and the rear opening are arranged adjacent to each other.
8. The woofer system according to claim 4, wherein the front opening is arranged close to one of ears, and the rear opening is arranged close to an other of the ears.
9. The woofer system according to claim 1, wherein the sound path element includes a right-side opening arranged close to a right ear of the listener who sits on the seat and a left-side opening arranged close to a left ear of the listener.
10. An acoustic system comprising: the woofer system according to claim 1; and a loudspeaker that is attached to the seat in a vicinity of the ear of the listener who sits on the seat, and reproduces an acoustic signal in a mid-to-high frequency sound range that is a range including frequencies higher than frequencies at which the woofer reproduces an acoustic signal.
11. A seat comprising: a woofer; and a sound path element that forms a sound path that leads a sound output by the woofer, wherein the sound path element includes an opening arranged close to an ear of a listener who sits on the seat.
12. A mobile object comprising: one or more of woofer systems each of which is the woofer system according to claim 1.